

Debt Service Flexibility and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Service Flexibility (DSF), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DSF achieves an annualized gross (net) Sharpe ratio of 0.45 (0.40), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (24) bps/month with a t-statistic of 2.65 (2.84), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals) is 24 bps/month with a t-statistic of 2.71.

1 Introduction

Production-based asset pricing models predict that firms' operational flexibility and financial constraints shape their exposure to systematic risk through investment frictions and adjustment costs. While extensive literature documents how real frictions affect the cross-section of expected returns, the role of debt service flexibility in determining firms' marginal costs of production and investment remains underexplored. We address this gap by examining how debt service flexibility (DSF) predicts stock returns through its impact on firms' production decisions and their ability to respond to productivity shocks.

In production economies, firms with lower debt service flexibility face higher marginal costs of adjusting capital in response to productivity shocks, creating systematic variation in conditional risk premia [Cochrane and Piazzesi \(2005\)](#). When debt obligations are inflexible, firms experience amplified exposure to aggregate productivity shocks through the financial accelerator mechanism, as rigid debt service requirements constrain optimal investment timing and increase adjustment costs [Zhang \(2005\)](#). This friction becomes particularly binding during negative productivity shocks, when firms with inflexible debt structures cannot efficiently scale down operations or defer capital expenditures.

The production-based framework predicts that debt service flexibility acts as a hedge against investment frictions by reducing the convexity of adjustment costs [Cooper and Haltiwanger \(2006\)](#). Firms with more flexible debt arrangements maintain option value in their production decisions, allowing them to time investment optimally when productivity shocks arrive. This operational flexibility reduces their exposure to systematic risk factors that price the cross-section of returns through the stochastic discount factor [Berk et al. \(1999\)](#). Consequently, firms with higher debt service flexibility should command lower risk premia in equilibrium.

Furthermore, debt service flexibility directly maps to the shadow cost of external

finance in dynamic production models [Gomes and Schmid \(2021\)](#). When firms face binding financial constraints, inflexible debt service requirements increase the wedge between the marginal product of capital and the cost of investment, amplifying their exposure to aggregate investment shocks. This mechanism generates cross-sectional variation in expected returns that reflects heterogeneity in firms' production technologies and their ability to substitute between internal and external financing sources [Livdan et al. \(2009\)](#).

The DSF strategy generated a value-weighted long/short return of 23 basis points per month with a t-statistic of 2.71, yielding an annualized Sharpe ratio of 0.45. The strategy's alpha relative to the Fama-French six-factor model reached 23 basis points per month with a t-statistic of 2.65, demonstrating robust performance after controlling for established risk factors. Notably, the strategy maintained significant alphas ranging from 21 to 28 basis points across all standard factor models, with the lowest alpha of 21 basis points occurring relative to the four-factor model.

The economic magnitude of the DSF effect remained substantial among large-cap stocks, where the strategy earned 37 basis points per month with a t-statistic of 3.36 within the largest size quintile. Transaction cost analysis revealed that the strategy's net Sharpe ratio of 0.40 ranked in the 95th percentile among 212 documented anomalies, while net returns after accounting for implementation costs averaged 21 basis points per month. The strategy significantly expanded the mean-variance efficient frontier in twenty out of twenty-five alternative portfolio construction methodologies tested.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the DSF strategy retained an alpha of 24 basis points per month with a t-statistic of 2.71. The strategy's performance proved robust across alternative sorting methodologies, with twenty-four out of twenty-five gross alphas exhibiting t-statistics exceeding two. These results confirm that debt service flexi-

bility captures a distinct dimension of systematic risk related to production-based frictions not subsumed by existing factor models or related predictors.

This paper extends the production-based asset pricing literature by documenting how debt service flexibility shapes cross-sectional return predictability through investment frictions and adjustment costs. While [Whited and Wu \(2006\)](#) establish that financial constraints affect expected returns through the investment channel, and [Li et al. \(2009\)](#) demonstrate the role of operating leverage in production-based models, we provide the first comprehensive evidence linking debt service flexibility to systematic risk premia through marginal production costs. Our findings complement [Ozdogli and Rubio-Ramirez \(2020\)](#) by showing that debt structure flexibility represents a distinct source of cross-sectional variation beyond traditional leverage measures.

Methodologically, we advance the anomaly testing framework by implementing the comprehensive protocol of [Novy-Marx and Velikov \(2023\)](#), ensuring our results withstand stringent robustness tests including transaction costs and alternative portfolio constructions. Unlike [Hou et al. \(2015\)](#), who focus on profitability and investment factors derived from q-theory, our approach directly measures operational flexibility through debt service arrangements, providing a more granular view of how financial frictions affect production decisions. The DSF signal’s orthogonality to existing factors, particularly after controlling for accruals and book-to-market variations, establishes it as a novel predictor that captures previously unexplained variation in the cross-section.

Our results have broader implications for understanding how production-based mechanisms generate asset pricing anomalies in equilibrium. The persistence of the DSF premium after controlling for size, momentum, and profitability factors suggests that markets incompletely incorporate information about firms’ operational flexibility into prices, consistent with models where investors face information pro-

cessing constraints about complex production technologies [Van Nieuwerburgh and Veldkamp \(2009\)](#). These findings inform both theoretical models of production-based asset pricing and practical portfolio construction by identifying a robust source of cross-sectional return predictability rooted in fundamental economic mechanisms.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial statement items that enable consistent measurement across firms and over time. Our sample includes all non-financial firms with available data for the required variables, spanning several decades of annual observations. Debt Service Flexibility signal combines two key financial statement items. Assets and liabilities other net change (AOLOCH) represents the net change in balance sheet items not separately disclosed elsewhere in the statement of cash flows, capturing residual adjustments to assets and liabilities that reconcile net income to operating cash flow. This item reflects various non-cash adjustments and reclassifications that affect the balance sheet during the reporting period. Interest expense (XINT) measures the periodic cost of debt financing, representing the total interest charges incurred on all outstanding debt obligations during the fiscal year. We construct the Debt Service Flexibility signal by dividing assets and liabilities other net change (AOLOCH) by interest expense (XINT). This ratio captures the relationship between balance sheet adjustments and the firm’s debt servicing obligations, potentially indicating the firm’s flexibility in managing working capital and other balance sheet items relative to its fixed interest commitments. We calculate this measure annually using fiscal year-end values and require non-missing, non-zero values for interest expense to ensure meaningful ratios. Firms with zero or negative interest expense are

excluded from the analysis, as are observations with missing values for either component. The resulting signal provides a scaled measure that facilitates cross-sectional comparisons across firms of different sizes and capital structures.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DSF signal. Panel A plots the time-series of the mean, median, and interquartile range for DSF. On average, the cross-sectional mean (median) DSF is 1.90 (-0.03) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input DSF data. The signal’s interquartile range spans -1.70 to 1.48. Panel B of Figure 1 plots the time-series of the coverage of the DSF signal for the CRSP universe. On average, the DSF signal is available for 5.82% of CRSP names, which on average make up 7.25% of total market capitalization.

4 Does DSF predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DSF using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DSF portfolio and sells the low DSF portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DSF strategy earns an average return of 0.23% per month with a t-statistic of 2.71. The annualized Sharpe ratio of the strategy is 0.45. The alphas range from 0.21% to 0.28% per month and

have t-statistics exceeding 2.47 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.08, with a t-statistic of 3.99 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 522 stocks and an average market capitalization of at least \$2,381 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 3.05. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas

measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -2-37bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DSF trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the DSF strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DSF, as well as average returns and alphas for long/short trading DSF strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DSF strategy achieves an average return of 37 bps/month with a t-statistic of 3.36. Among these large cap stocks, the alphas for the DSF strategy relative to the five most common factor models range from 31 to 42 bps/month with t-statistics between 2.85 and 3.74.

5 How does DSF perform relative to the zoo?

Figure 2 puts the performance of DSF in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DSF strategy falls in the distribution. The DSF strategy’s gross (net) Sharpe ratio of 0.45 (0.40) is greater than 85% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DSF strategy (red line).² Ignoring trading costs, a \$1 invested in the DSF strategy would have yielded \$1.83 which ranks the DSF strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DSF strategy would have yielded \$1.55 which ranks the DSF strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DSF relative to those. Panel A shows that the DSF strategy gross alphas fall between the 50 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DSF strategy has a positive net generalized alpha for five out of the five factor models. In these cases DSF ranks between the 77 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does DSF add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DSF with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DSF or at least to weaken the power DSF has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DSF conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DSF. Stocks are finally grouped into five DSF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

DSF trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DSF and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DSF signal in these Fama-MacBeth regressions exceed 1.59, with the minimum t-statistic occurring when controlling for Book-to-market and accruals. Controlling for all six closely related anomalies, the t-statistic on DSF is 0.42.

Similarly, Table 5 reports results from spanning tests that regress returns to the DSF strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DSF strategy earns alphas that range from 22-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.54, which is achieved when controlling for Book-to-market and accruals. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DSF trading strategy achieves an alpha of 24bps/month with a t-statistic of 2.71.

7 Does DSF add relative to the whole zoo?

Finally, we can ask how much adding DSF to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the DSF signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes DSF grows to \$31.51.

8 Conclusion

This study provides robust evidence that Debt Service Flexibility (DSF) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DSF-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on DSF delivers impressive performance metrics, with an annualized gross Sharpe ratio of 0.45 and a net Sharpe ratio of 0.40 after accounting for transaction costs. More importantly, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (23-24 bps/month with t-statistics exceeding 2.65) underscores its distinct information content. The persistence of these results even after controlling for six closely related strategies from the factor zoo—including changes in current operating liabilities, book-to-market and accruals, and total accruals—further validates DSF as an independent predictor of returns rather than a mere repackaging of existing anomalies.

ization on CRSP in the period for which DSF is available.

From a practical perspective, these findings have important implications for both investors and corporate managers. For investors, DSF offers a viable strategy for portfolio construction that appears robust to transaction costs, as evidenced by the minimal degradation between gross and net returns. For corporate managers, the results highlight the market’s appreciation of financial flexibility in debt servicing, suggesting that maintaining such flexibility can positively impact firm valuation.

However, this study is not without limitations. First, while we follow the Novy-Marx and Velikov (2023) protocol, the specific economic mechanisms driving the DSF premium warrant further investigation. Second, the analysis is confined to U.S. equity markets, and the generalizability of these findings to international markets remains unexplored. Third, the stability of the DSF signal across different market regimes and its performance during periods of financial stress require additional scrutiny.

Future research should address these limitations by examining the underlying economic channels through which debt service flexibility affects stock returns. Investigating whether the DSF premium varies with macroeconomic conditions, credit market cycles, or monetary policy regimes could provide valuable insights into its risk-based versus mispricing origins. Additionally, extending the analysis to international markets would help establish whether DSF represents a global phenomenon or is specific to U.S. market structures. Finally, exploring the interaction between DSF and corporate financial policies, such as capital structure decisions and investment behavior, could enhance our understanding of how firms optimize their debt service flexibility and how markets price this optimization.

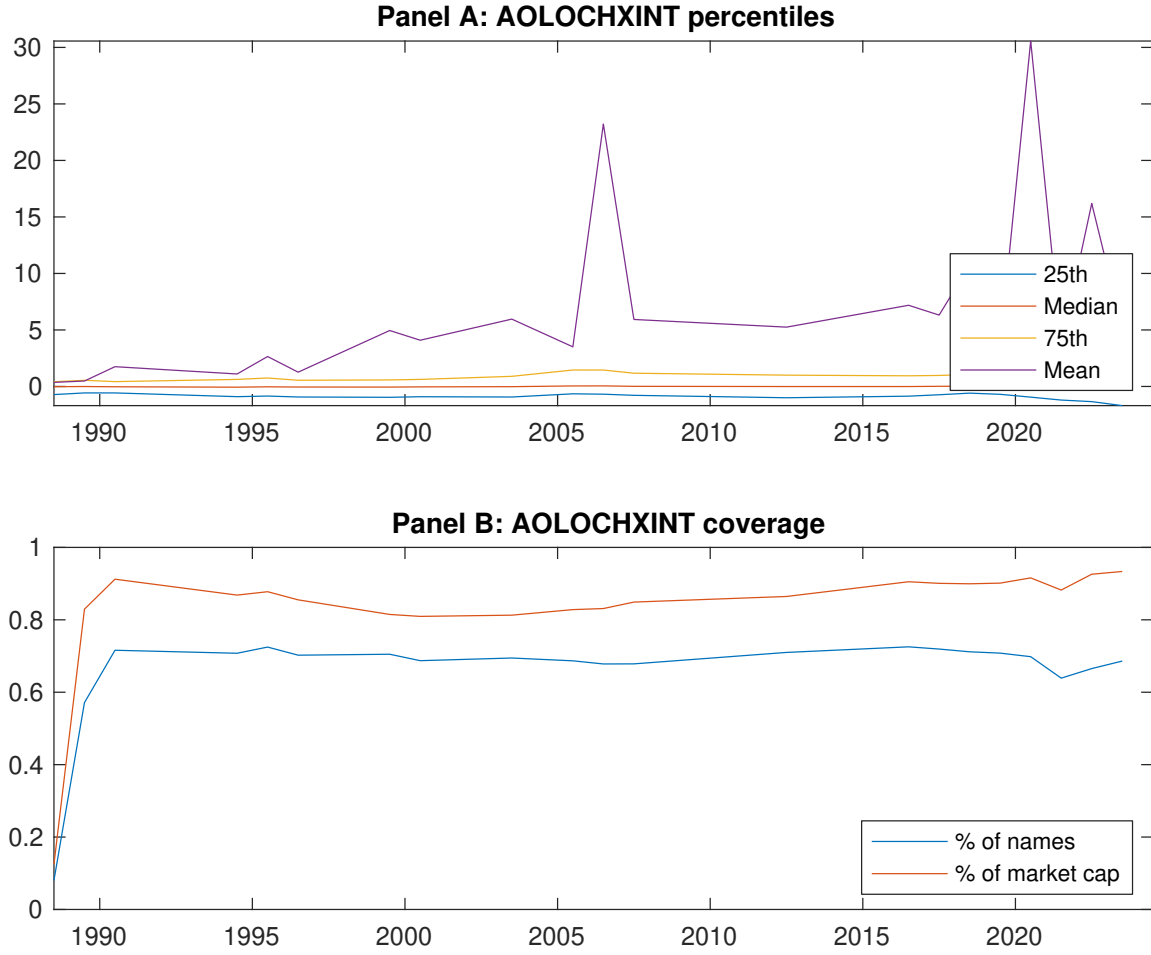


Figure 1: Times series of DSF percentiles and coverage.
This figure plots descriptive statistics for DSF. Panel A shows cross-sectional percentiles of DSF over the sample. Panel B plots the monthly coverage of DSF relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DSF. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on DSF-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.62 [2.78]	0.79 [3.79]	0.70 [3.57]	0.72 [3.54]	0.86 [3.95]	0.23 [2.71]
α_{CAPM}	-0.14 [-2.45]	0.09 [1.35]	0.06 [0.78]	0.03 [0.52]	0.11 [2.04]	0.26 [2.93]
α_{FF3}	-0.13 [-2.24]	0.06 [0.96]	0.01 [0.20]	0.01 [0.12]	0.14 [2.73]	0.27 [3.12]
α_{FF4}	-0.09 [-1.52]	0.06 [0.91]	0.02 [0.26]	0.02 [0.26]	0.13 [2.44]	0.21 [2.47]
α_{FF5}	-0.11 [-1.90]	-0.04 [-0.59]	-0.15 [-2.36]	-0.10 [-1.65]	0.17 [3.25]	0.28 [3.19]
α_{FF6}	-0.08 [-1.32]	-0.03 [-0.51]	-0.14 [-2.14]	-0.08 [-1.41]	0.16 [2.96]	0.23 [2.65]
Panel B: Fama and French (2018) 6-factor model loadings for DSF-sorted portfolios						
β_{MKT}	1.01 [68.74]	0.99 [60.59]	0.94 [58.60]	0.99 [67.91]	1.00 [75.43]	-0.02 [-0.78]
β_{SMB}	0.02 [0.94]	0.00 [0.15]	-0.00 [-0.07]	-0.03 [-1.39]	-0.05 [-2.64]	-0.07 [-2.21]
β_{HML}	-0.09 [-3.59]	0.03 [0.95]	0.04 [1.50]	-0.01 [-0.24]	-0.09 [-3.93]	0.00 [0.05]
β_{RMW}	-0.05 [-1.94]	0.13 [4.49]	0.28 [9.47]	0.17 [6.39]	-0.03 [-1.11]	0.03 [0.64]
β_{CMA}	0.06 [1.67]	0.19 [4.61]	0.21 [5.04]	0.14 [3.74]	-0.10 [-2.92]	-0.16 [-2.87]
β_{UMD}	-0.06 [-4.26]	-0.01 [-0.57]	-0.02 [-1.46]	-0.02 [-1.59]	0.02 [1.89]	0.08 [3.99]
Panel C: Average number of firms (n) and market capitalization (me)						
n	832	563	522	570	784	
me (\$10 ⁶)	2927	2434	2381	2893	4206	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DSF strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23 [2.71]	0.26 [2.93]	0.27 [3.12]	0.21 [2.47]	0.28 [3.19]	0.23 [2.65]
Quintile	NYSE	EW	0.19 [3.05]	0.21 [3.29]	0.21 [3.36]	0.14 [2.29]	0.20 [3.05]	0.14 [2.21]
Quintile	Name	VW	0.27 [2.59]	0.29 [2.77]	0.30 [2.94]	0.21 [2.10]	0.27 [2.56]	0.20 [1.90]
Quintile	Cap	VW	0.28 [2.87]	0.29 [2.92]	0.31 [3.19]	0.25 [2.53]	0.36 [3.54]	0.30 [2.97]
Decile	NYSE	VW	0.40 [3.03]	0.41 [3.08]	0.44 [3.31]	0.30 [2.32]	0.44 [3.18]	0.32 [2.39]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.44]	0.25 [2.90]	0.26 [3.07]	0.23 [2.69]	0.28 [3.17]	0.24 [2.84]
Quintile	NYSE	EW	-0.02 [-0.23]	0.01 [0.10]				
Quintile	Name	VW	0.24 [2.34]	0.29 [2.81]	0.30 [2.96]	0.25 [2.47]	0.28 [2.71]	0.23 [2.29]
Quintile	Cap	VW	0.26 [2.64]	0.29 [2.98]	0.31 [3.22]	0.27 [2.83]	0.35 [3.58]	0.31 [3.23]
Decile	NYSE	VW	0.37 [2.80]	0.38 [2.83]	0.40 [3.03]	0.32 [2.45]	0.40 [2.96]	0.33 [2.49]

Table 3: Conditional sort on size and DSF

This table presents results for conditional double sorts on size and DSF. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DSF. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DSF and short stocks with low DSF .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DSF Quintiles					DSF Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.55 [1.64]	0.80 [2.43]	0.72 [1.99]	0.82 [2.49]	0.75 [2.25]	0.20 [2.04]	0.18 [1.78]	0.16 [1.62]	0.14 [1.37]	0.19 [1.80]	0.17 [1.58]
	(2)	0.70 [2.26]	0.74 [2.49]	0.86 [2.95]	0.75 [2.64]	0.74 [2.43]	0.04 [0.39]	0.09 [0.77]	0.11 [1.03]	0.06 [0.53]	0.11 [0.96]	0.06 [0.54]
	(3)	0.74 [2.72]	0.72 [2.70]	0.81 [3.02]	0.83 [3.20]	0.81 [2.93]	0.07 [0.64]	0.06 [0.52]	0.09 [0.81]	0.02 [0.21]	0.19 [1.75]	0.13 [1.21]
	(4)	0.82 [3.09]	0.72 [3.01]	0.70 [2.97]	0.83 [3.54]	0.90 [3.36]	0.08 [0.84]	0.07 [0.73]	0.10 [1.13]	0.05 [0.55]	0.17 [1.81]	0.12 [1.28]
	(5)	0.61 [2.76]	0.81 [4.08]	0.69 [3.53]	0.67 [3.31]	0.97 [4.39]	0.37 [3.36]	0.37 [3.32]	0.39 [3.54]	0.31 [2.85]	0.42 [3.74]	0.35 [3.15]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DSF Quintiles					DSF Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	344	343	340	341	343	36	38	38	37	38	
	(2)	108	108	108	108	108	72	74	74	74	74	
	(3)	76	77	76	77	77	131	133	134	134	133	
	(4)	67	66	67	66	67	295	303	300	308	304	
(5)	61	61	61	61	61	2517	2051	2131	2508	2902		

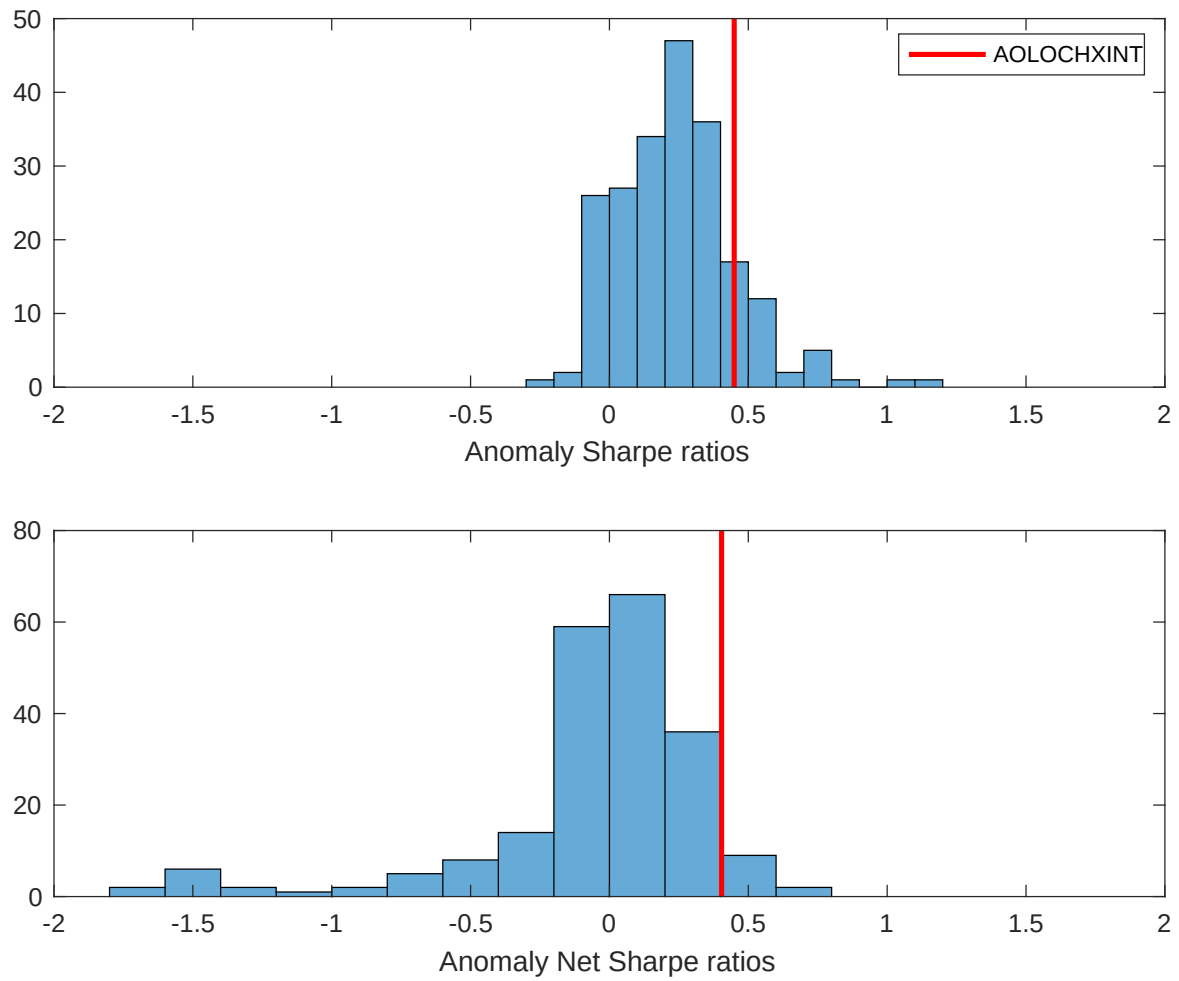


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DSF with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

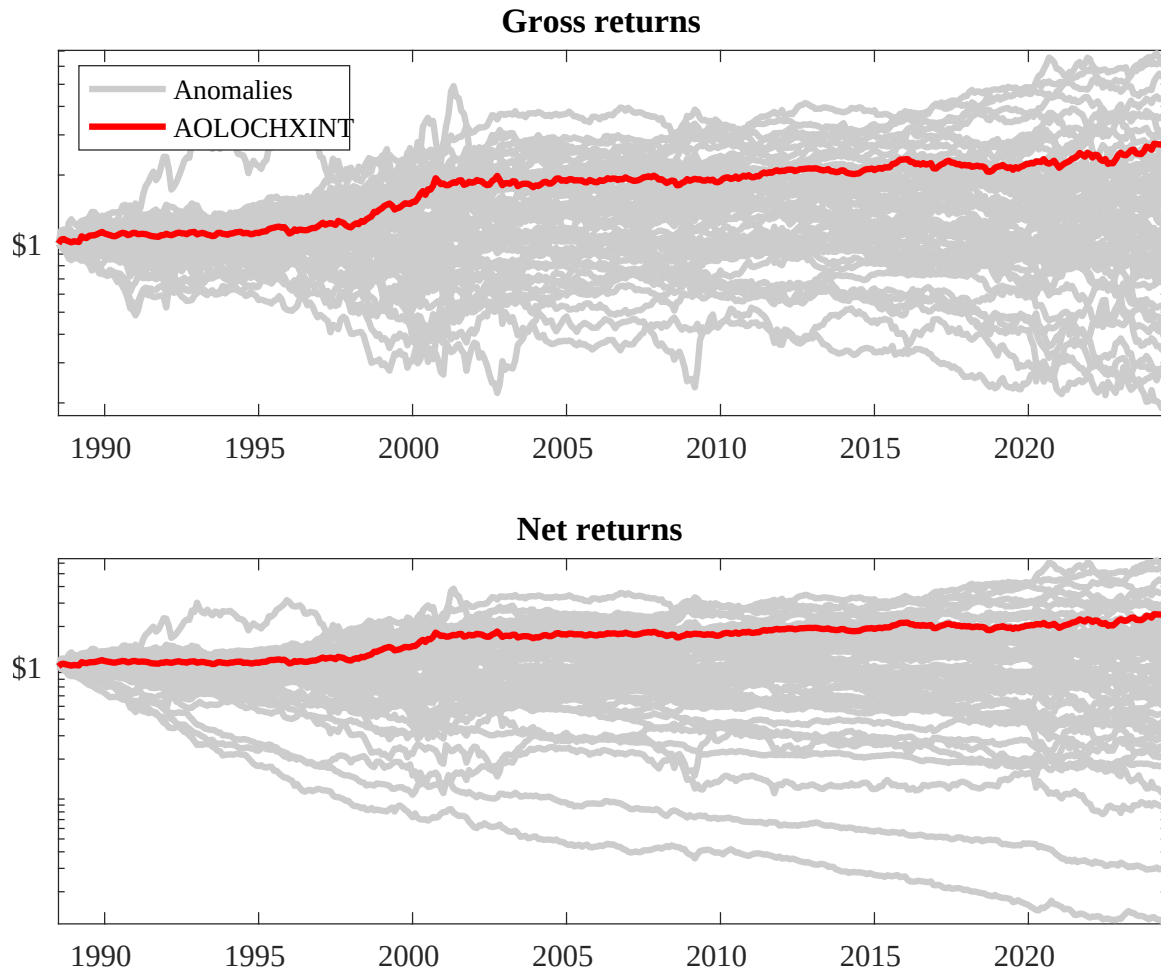
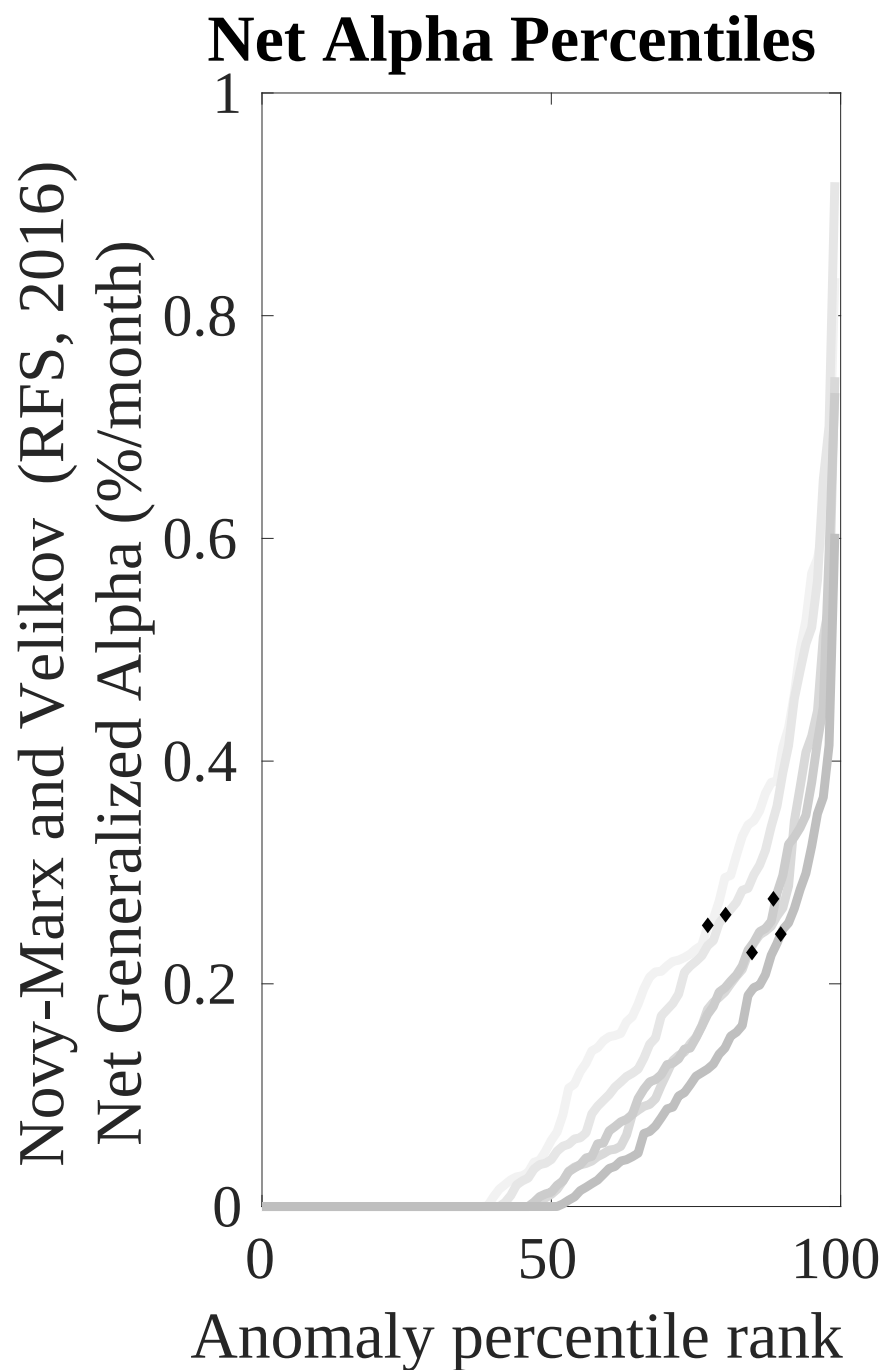
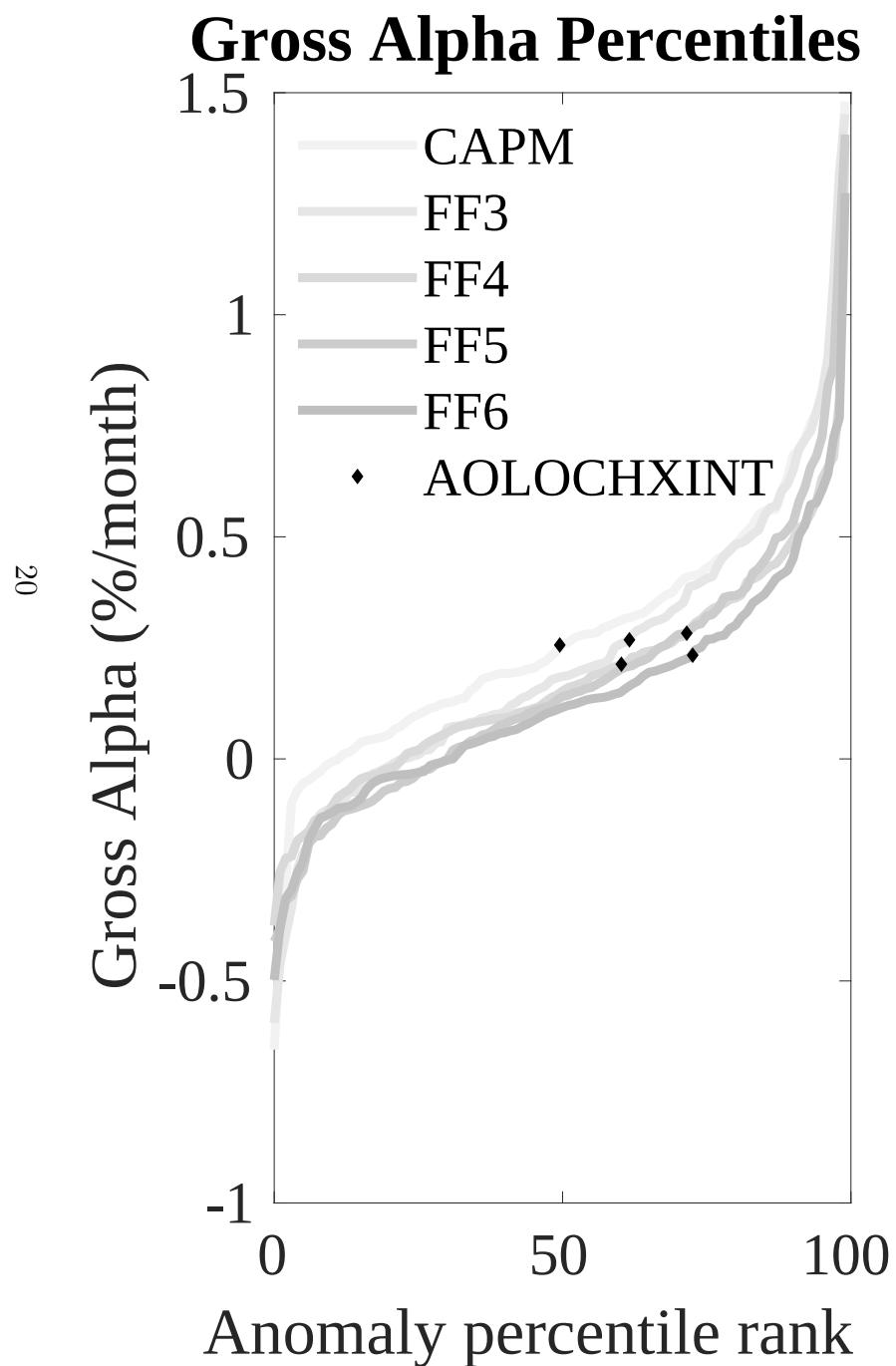


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DSF trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



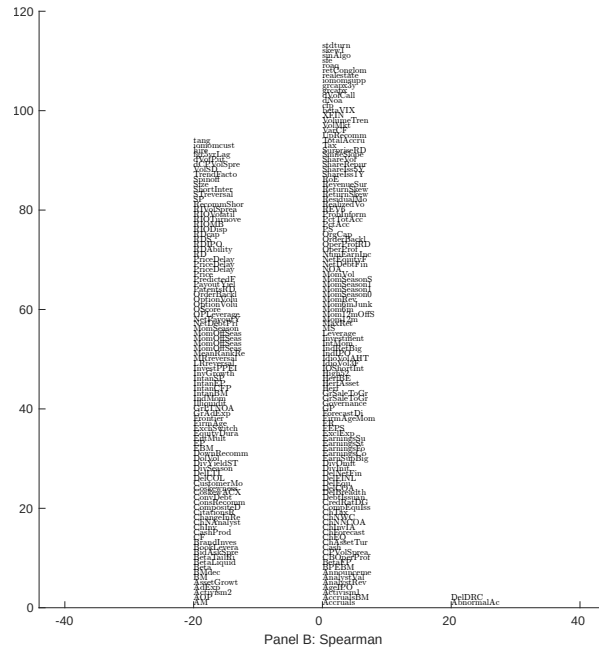
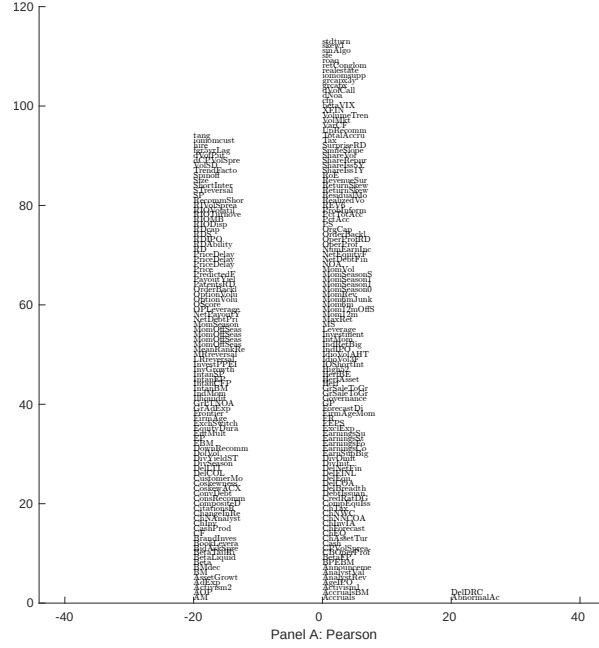


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with DSF. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

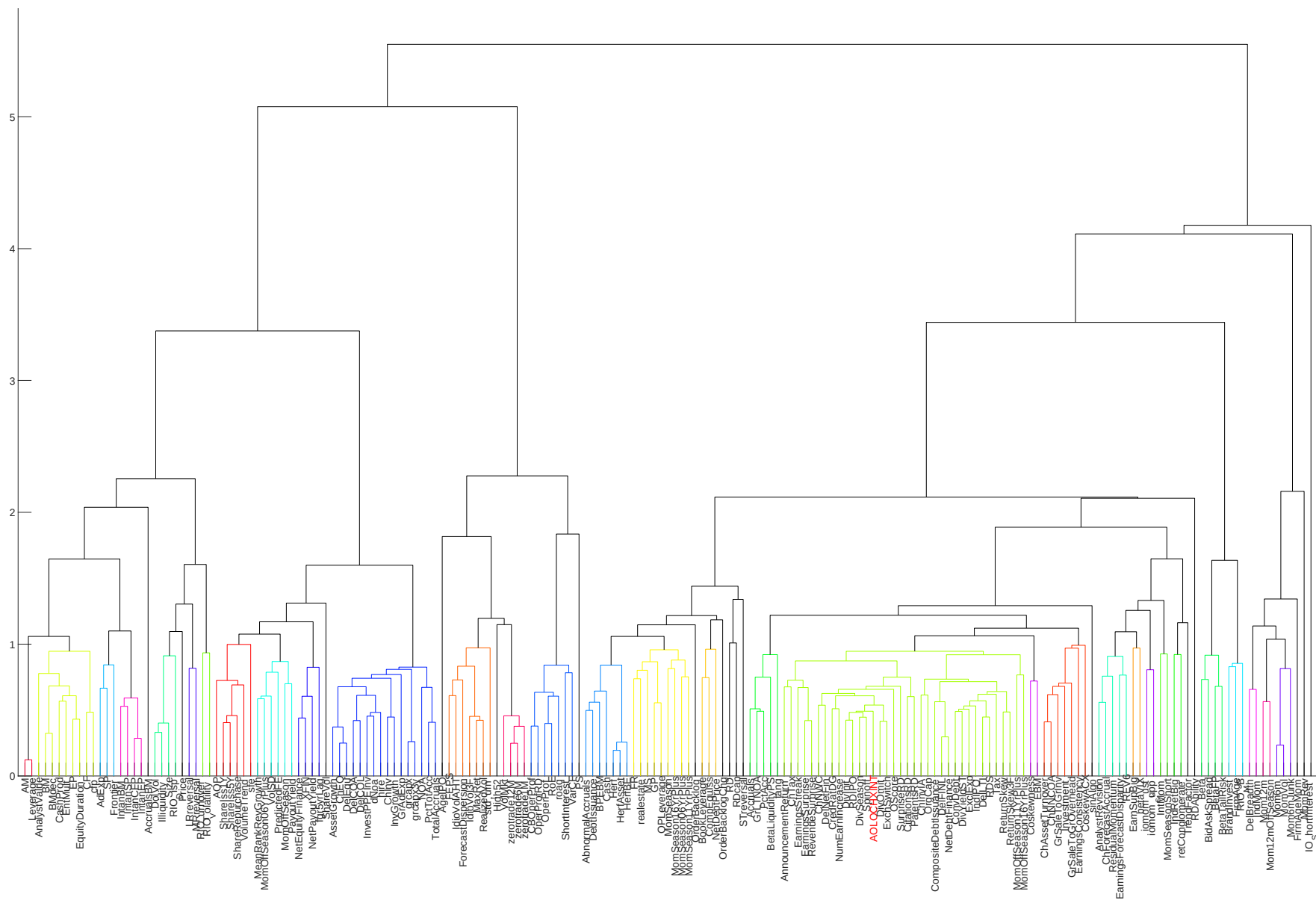


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

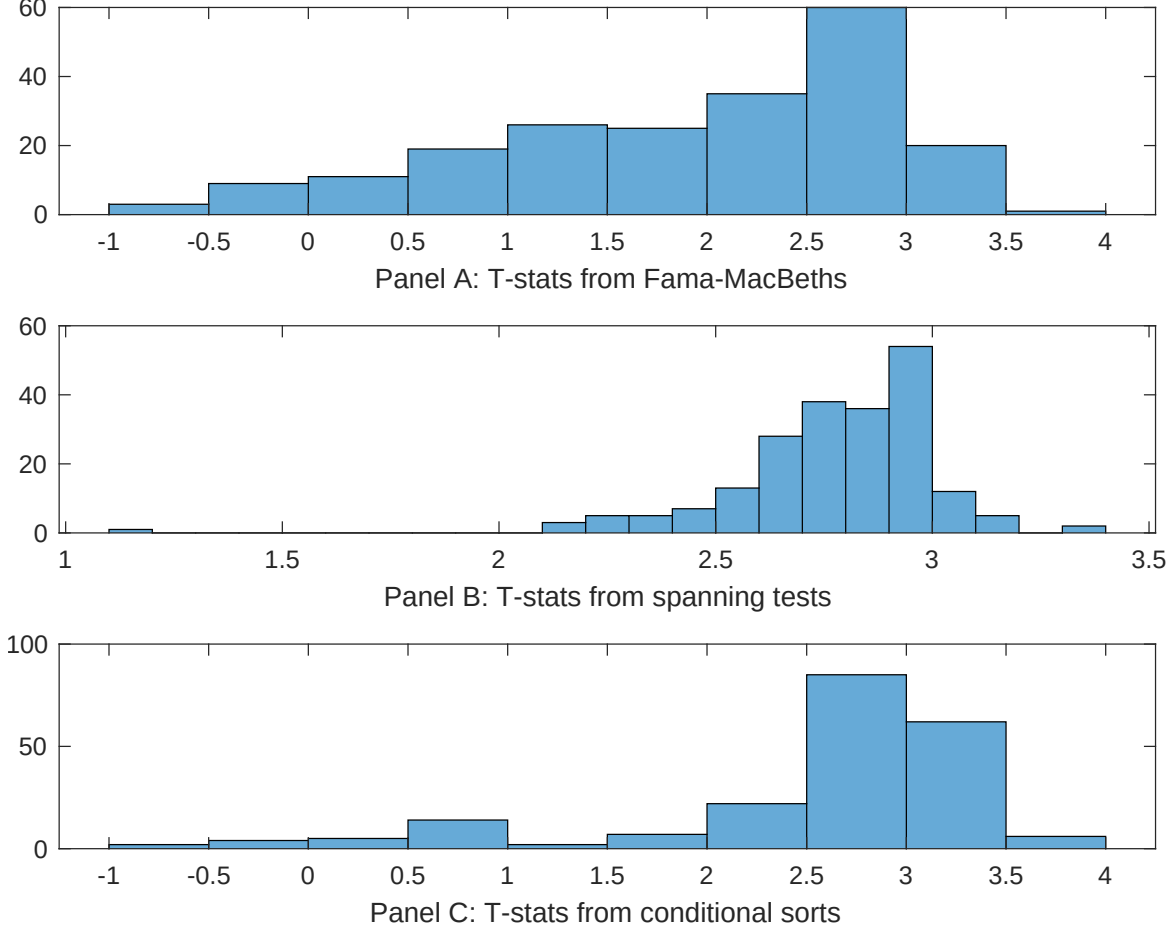


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DSF conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DSF. Stocks are finally grouped into five DSF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSF trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DSF. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11 [3.95]	0.66 [1.85]	0.11 [3.87]	0.11 [3.90]	0.11 [2.97]	0.11 [3.86]	0.24 [3.38]
DSF	0.35 [3.05]	0.80 [1.59]	0.25 [1.68]	0.31 [2.71]	0.36 [3.21]	0.26 [2.21]	0.28 [0.42]
Anomaly 1	0.19 [4.46]						1.00 [0.58]
Anomaly 2		0.14 [4.58]					-0.25 [-3.26]
Anomaly 3			0.44 [4.46]				0.20 [6.41]
Anomaly 4				0.14 [5.38]			0.20 [2.25]
Anomaly 5					0.45 [0.88]		-0.75 [-0.81]
Anomaly 6						0.47 [1.64]	0.67 [1.05]
# months	432	415	427	432	432	432	267
$\bar{R}^2(\%)$	0	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DSF trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DSF} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.22 [2.54]	0.26 [2.99]	0.26 [2.93]	0.26 [3.05]	0.26 [2.96]	0.26 [3.03]	0.24 [2.71]
Anomaly 1	-12.60 [-3.09]						-8.98 [-2.02]
Anomaly 2		-1.21 [-0.96]					-1.07 [-0.85]
Anomaly 3			-0.40 [-0.15]				0.77 [0.27]
Anomaly 4				17.50 [5.03]			15.42 [4.24]
Anomaly 5					0.34 [0.10]		-0.42 [-0.12]
Anomaly 6						-7.57 [-1.87]	-1.45 [-0.33]
mkt	-1.62 [-0.75]	-1.95 [-0.89]	-2.03 [-0.92]	-3.61 [-1.68]	-1.94 [-0.86]	-1.74 [-0.80]	-3.17 [-1.41]
smb	-7.48 [-2.39]	-5.70 [-1.78]	-6.07 [-1.79]	-6.38 [-2.09]	-6.63 [-1.51]	-6.89 [-2.19]	-6.80 [-1.59]
hml	7.07 [1.65]	2.40 [0.58]	1.19 [0.30]	3.45 [0.92]	0.86 [0.22]	0.17 [0.05]	8.39 [1.76]
rmw	2.44 [0.62]	1.36 [0.34]	1.90 [0.48]	1.98 [0.51]	2.33 [0.52]	0.52 [0.13]	1.05 [0.23]
cma	-10.72 [-1.81]	-16.70 [-2.97]	-17.09 [-3.03]	-15.25 [-2.81]	-17.38 [-3.10]	-12.10 [-1.95]	-9.37 [-1.48]
umd	7.23 [3.74]	7.10 [3.40]	7.55 [2.86]	3.34 [1.60]	8.07 [2.73]	7.20 [3.66]	2.89 [0.91]
# months	432	428	428	432	432	432	428
$\bar{R}^2(\%)$	10	8	8	13	8	9	13

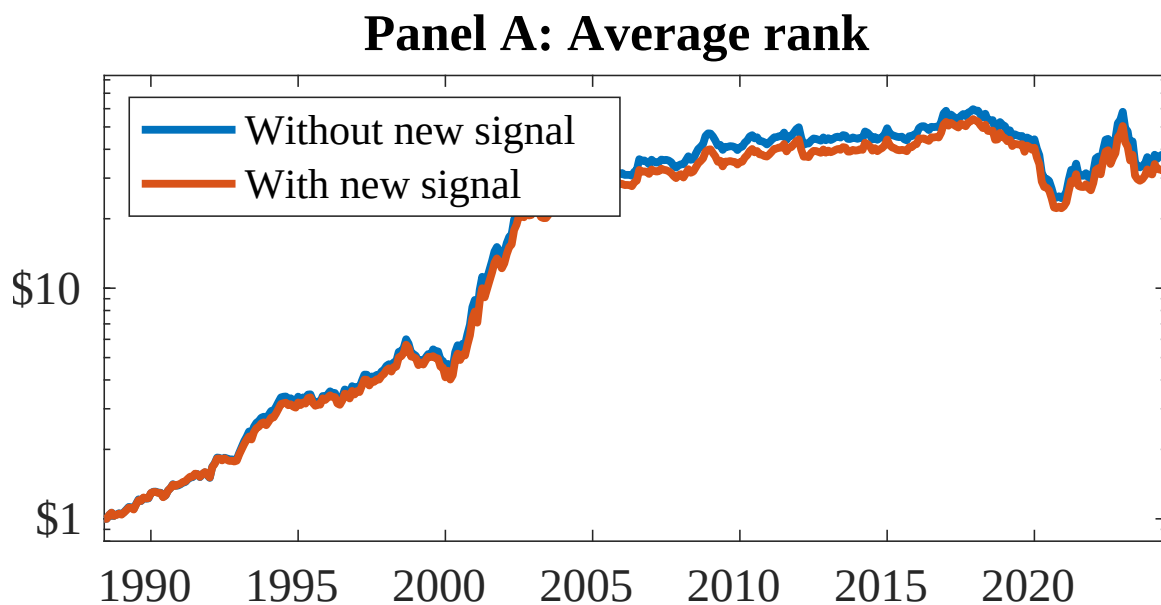


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as DSF. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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